Experiment 5

Fourier Series Analysis of Periodic Signals

OBJECTIVE 1:

To analyze Fourier series of continuous-time periodic signals and analyze their waveform characteristics.

Procedure

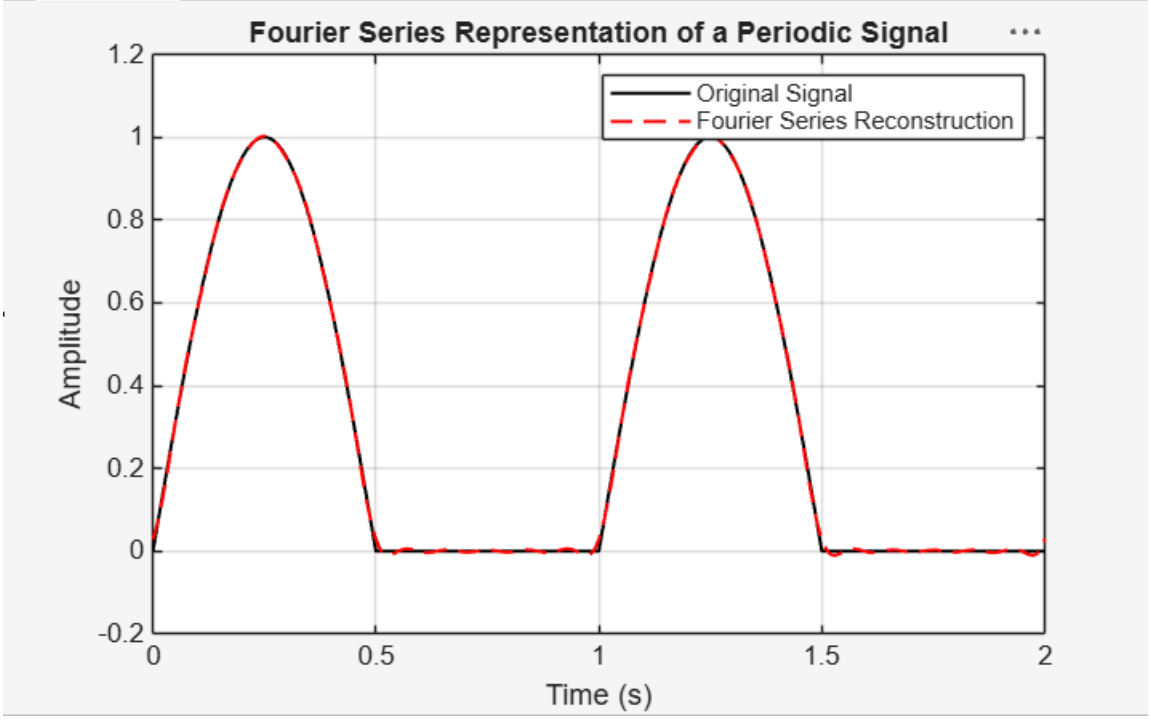
1. Define the time vector and fundamental frequency.
2. Generate the half-wave rectified sinusoidal periodic signal.
3. Compute the Fourier coefficients up to N = 10.
4. Construct the Fourier-series approximation.
5. Plot the original and reconstructed signals and compare them.

MATLAB Coding

```
clc; clear; close all;
%% 5(i) Analysis of Fourier Series Representation of Continuous-Time Periodic Signals
figure;
% Fourier Series Representation of a Periodic Signal
A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
a0 = (2/T)*trapz(t, x)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
```

```
an(k) = (2/T)*trapz(t, x.*cos(2*pi*k*f0*t));
bn(k) = (2/T)*trapz(t, x.*sin(2*pi*k*f0*t));
endxFS = a0*ones(size(t));
for k = 1:N
xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
+ bn(k)*sin(2*pi*k*f0*t);
end
plot(t,x,'k','LineWidth',1.2); hold on;
plot(t,xFS,'r--','LineWidth',1.2);
grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal','Fourier Series Reconstruction');
```

OUTPUT:



RESULT:

OBJECTIVE 2:

To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal

PROCEDURE:

1. Generate the periodic signal over two periods.

2. Compute Fourier coefficients from one period.

3. Construct the reconstructed signal using 10 harmonics.

4. Calculate the error signal.

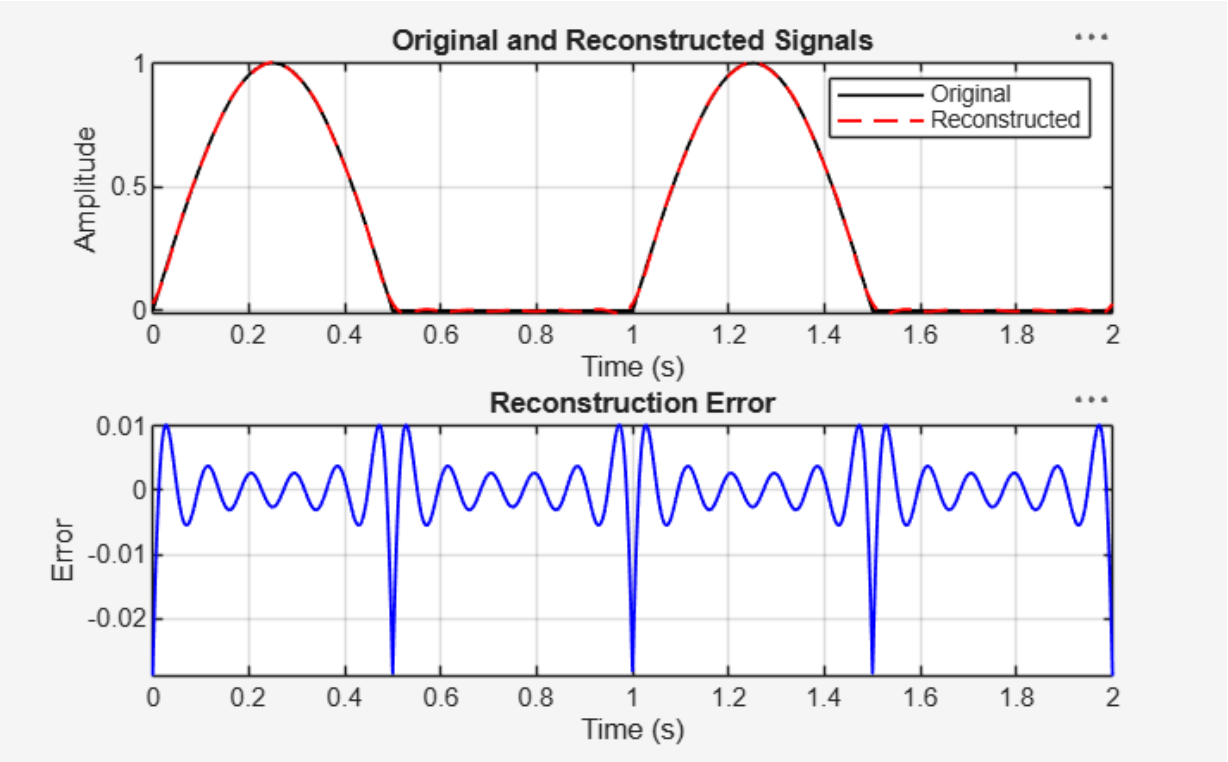
5. Plot the original, reconstructed and error signals.

MATLAB CODE:

```
clc; clear; close all;
%% 5(iv) Verification of Fourier Series Reconstruction and Analysis of Signal Characteristics
figure;
% Fourier Series Reconstruction and Error Analysis
f0 = 1;
T = 1/f0;
t = linspace(0,2*T,4000);
x = max(sin(2*pi*f0*t),0);
N = 10;
a0 = (2/T)*trapz(t(1:2000),x(1:2000));
an = zeros(1,N); bn = zeros(1,N);
tt = t(1:2000); xx = x(1:2000);
for k = 1:N
an(k) = (2/T)*trapz(tt,xx.*cos(2*pi*k*f0*tt));
bn(k) = (2/T)*trapz(tt,xx.*sin(2*pi*k*f0*tt));end
xr = a0/2*ones(size(t));
for k = 1:N
```

```
xr = xr + an(k)*cos(2*pi*k*f0*t) + bn(k)*sin(2*pi*k*f0*t);
end
err = x-xr;
subplot(2,1,1);
plot(t,x,'k',t,xr,'r--','LineWidth',1.1);
grid on; xlabel('Time (s)'); ylabel('Amplitude');
title('Original and Reconstructed Signals');
legend('Original','Reconstructed');
subplot(2,1,2);
plot(t,err,'b'); grid on;
xlabel('Time (s)'); ylabel('Error');
title('Reconstruction Error');
```

OUTPUT:



RESULT:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	
3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency, phase, and growth or decay. Explains the	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications . Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification .	**25**	**2.5**	

		reasoning clearly with mathematical conditions and waveform evidence.						
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	